

SWAPCELL · CALCULATION NOTE

SwapCell sizing and interface v0.3 calculations

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DOCUMENT CONTROL

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REVISION HISTORY

Version	Date	Author	Change
0.3	2026-10-01	Amish Chadha	Constructable design (SWC-DDR-003) mass, thermal and cost; budget treated as a value-engineering target
0.2	2026-09-25	Amish Chadha	Recommendations accepted by Amish (DDR-002)
0.1	2026-09-25	Amish Chadha	First TRL 3 sizing note (electrical, mass, thermal, charge, wake, charge-discharge, latch, connector, CAN, cost)

SwapCell sizing and interface v0.3 calculations

The 13S2P reference pack meets its voltage, charge time, mass, envelope and data requirements on paper, its parts cost is under the value-engineering target, and the three interface v0.3 additions are feasible with large margins. R3 is now met on paper as restated by Amish's decision of 2026-09-25 (SWC-DDR-002): 20 A from 25 °C in an open-air mount, with temperature derating in PACK_LIMITS elsewhere (about 18.5 A enclosed, about 14.0 A from 45 °C ambient). R9 (cycle life) stays **at risk**, and R2 is at risk only at the datasheet minimum cell capacity. No requirement is shown to be not met. Connector, sealing, protections and latch retention (R7, R8, R10, R11, R15) cannot be verified until hardware exists, which is TRL 4 work and on hold by Amish's instruction.

Every number in this note is printed by `docs/04-calcs/sizing.py` (run from the repo root: `python docs/04-calcs/sizing.py`), which also writes `docs/04-calcs/results.csv`. All values are first-principles estimates; nothing here is measured.

■ 1. Assumptions

Table 1. Inputs. All are assumptions for a paper design.

Input	Value	Basis
Cell	21700, 5.0 Ah nominal, 4.85 Ah minimum, 3.6 V nominal, 3.0 to 4.2 V, 69 g, 12 mΩ DC, 25 A continuous	Typical high-power 5 Ah 21700 class; the model is chosen at supplier selection
Cell specific heat	950 J/(kg K)	Typical for cylindrical NMC cells
Configuration	13S2P, 26 cells	Decided, SWC-DDR-001 item 1
Resistance outside the cells	32 mΩ	Links, fuse wires, FETs, shunt, main fuse, connector
Resistance rise late in discharge and with age	x 1.25 on cell resistance	Sensitivity case
Fleet-mode capacity	90 % of standard	Typical at 4.1 V per cell
Tray	1.5 mm 5052 aluminium, 2.70 g/cm ³ , 900 J/(kg K), with 8 mm inward front flanges	Decided, SWC-DDR-001 item 2; flanges SWC-DDR-003
Lid	3 mm flame-retardant polymer, 1.20 g/cm ³	PC/ABS class
External heat loss	9 W/(m ² K)	Still air, natural convection plus radiation
Cell core to tray	0.10 K/W	Holders and thermal pads
Charger	5 A CC-CV, 90 % efficient; CC to 85 %, then 0.6 h CV	Certified unit (decided)
Cell charge efficiency, motor and controller efficiency	95 %, 80 %	Typical
Contact insertion forces	12 N per power contact, 1.5 N per signal contact, 8 N latch detent	Assumed until the contact parts are chosen (family decided: custom keyed shroud, SWC-DDR-002)
Vibration and shock for latch class V1	8 g peak sine; 25 g, 11 ms half-sine	UN 38.3 test T3 peak for batteries under 12 kg; 25 g curb-strike level, decided by Amish 2026-09-25 (SWC-DDR-002)
Receiver design mass	3.5 kg	R5 limit, not the 3.03 kg pack
Sleep current	100 μA	Target for a BMS asleep with the INTERLOCK comparator armed
INTERLOCK sensing	100 kΩ pull-up from a 3.3 V sleep rail; 10 kΩ coding resistor in the receiver	Interface v0.3 item W
Host input capacitance, pre-charge resistor	1,000 μF, 100 Ω	Typical 48 V controller
CAN frame	135 bits worst case (11-bit ID, 8 data bytes, bit stuffing)	CAN 2.0A frame on a 2.0B-capable bus
Use	2 swaps per day for 5 years	R10 basis

■ 2. Electrical

The pack gives 46.8 V nominal (39.0 to 54.6 V) and 10.0 Ah. At 0.2C the usable energy is about 466 Wh with nominal cells and about 452 Wh with cells at their 4.85 Ah minimum, which still clears the 450 Wh target but gives only 9.7 Ah against the 10 Ah target. In fleet mode (4.1 V per cell) the pack stores about 419 Wh; R2 applies to standard mode.

Pack resistance is about 110 mΩ (13 groups at 6 mΩ plus 32 mΩ). At 20 A each cell carries 10.0 A and the pack sags about 2.2 V; at 35 A each cell carries 17.5 A (rated 25 A) and the pack sags about 3.9 V.

For protections (R11): a bolted short at 54.6 V draws about 496 A, so a 500 μs trip lets through about 123 A²s, which sets the FET and main-fuse coordination. A cell with an internal short draws about 300 A from its parallel neighbour, which is the current the cell fuse wire must interrupt while carrying 17.5 A without damage. A 100 Ω pre-charge resistor charges 1,000 μF of host capacitance with a 100 ms time constant, reaching 99 % in about 460 ms and dissipating about 1.49 J.

■ 3. Mass and size

Table 2. Mass budget.

Part	Mass (kg)	Basis
Cells	1.79	26 x 69 g
Tray	0.43	1.5 mm Al, back, two sides, two ends and the front flanges (computed from the envelope)
Lid	0.11	3 mm over 340 x 90 mm
Interconnects, holders, insulation	0.18	Estimate
BMS board	0.12	Estimate
Plug	0.06	Estimate
Handle and latch pawl	0.13	Model volumes: steel pawl 11.4 cm ³ , printed handle 30 cm ³
Latch housing, release slider, thumb button and springs	0.11	Added for construction (SWC-DDR-003)
Seals, fasteners, wake button	0.10	Estimate; press-in nuts, screws, inserts and rivets added (SWC-DDR-003)
Total	3.03 (6.7 lb)	R5 limit 3.5 kg

The body is 340 x 90 x 80 mm (2.45 L), giving about 154 Wh/kg and 191 Wh/L. The handle adds 35 mm and the plug 18 mm, so the overall length is 393 mm against the 400 mm limit. The parametric model (`cad/src/model.py`) reproduces these: pack 90 x 86 x 393 mm including the latch pawl, which stands 6 mm proud, and body 90 x 80 x 340 mm with the lid, gasket, wake button and screw heads flush.

■ 4. Thermal (R3)

A lumped model with heat loss replaces the TRL 2 adiabatic estimate. The pack heat capacity is about 2.29 kJ/K, the outer area 0.130 m², the loss conductance UA about 1.17 W/K and the thermal time constant about 33 min, close to the 30 min of a full discharge at 20 A. Cell hot-spot temperature is ambient plus the lumped rise, $\Delta T = (Q/UA)(1 - e^{-(t/\tau)})$, plus the core-to-tray drop.

Table 3. Cell hot spot at the end of a full 20 A discharge.

Case	Heat	Open air from 25 °C	Enclosed (adiabatic) from 25 °C	Open air from 45 °C
Base resistance	44 W	51 °C	63 °C	71 °C
Resistance x 1.25	52 W	56 °C	70 °C	76 °C

The 20 A rating holds in open air from 25 °C with 4 to 9 K margin, but not in an enclosed vehicle mount or from 45 °C ambient. The continuous current that holds 60 °C over a full discharge is about 18.6 A in an enclosed (adiabatic) mount from 25 °C and about 14.0 A in open air from 45 °C. At a typical e-bike 10 A the pack makes only about 11 W.

Amish decided on 2026-09-25 (SWC-DDR-002) to keep the 20 A rating at 25 °C with temperature derating, and R3 is restated to match: 20 A in an open-air mount from 25 °C, and in any other case the pack lowers the allowed discharge current in PACK_LIMITS so the cells stay below 60 °C. The derating rule in SWC-PRC-001 v0.4 (full current to 50 °C cell temperature, falling linearly to 5 A at 60 °C) brings the pack to about the currents above without host action. R3 as restated is **met on paper**; it still needs a load test, which is TRL 4 work and on hold.

■ 5. Charging and energy chain (R4)

At 5 A the CC phase to 85 % takes 1.70 h and the full charge about 2.3 h; 0 to 80 % takes 1.6 h and 20 to 80 % takes 1.2 h. R4 (3 h full, 2 h to 80 %) is met. The dock draws about 303 W from the grid in CC.

Per full cycle: about 547 Wh from grid or solar, 493 Wh out of the charger, 468 Wh stored, 457 Wh at the pack terminals (10 A mean) and about 366 Wh at the wheel. That gives about 38 to 57 km for an e-bike at 12 to 8 Wh/km and about 18 to 30 km for a loaded cargo trike at 25 to 15 Wh/km. Figure 2 of SWC-PRC-001 uses these values.

■ 6. Interface v0.3 additions

Wake without CAN or a wake supply (item W, R13). In sleep the BMS holds INTERLOCK at 3.3 V through 100 kΩ. A v0.3 receiver loops INTERLOCK to SGND through 10 kΩ, pulling the node to about 0.30 V and drawing about 30 μA from the pack, so the host needs no supply of its own. A bridged short reads 0.00 V and an open loop 3.30 V; both are rejected, so a coin or wet debris across the contacts wakes the BMS but never enables the output. At 100 μA the sleep drain is about 0.72 % of capacity per month, so a half-charged pack takes about 5.7 years to sleep itself flat. R13 is met on paper.

Charge while discharging (item C, R14). A PowerBox station host with 200 W of solar in and a 20 W router load puts a net 3.8 A into the pack, inside the 5.0 A (0.5C) standard charge limit. StepGen-type regeneration of 5 to 10 W is far smaller. R14 is met on paper.

Latch class V1 (item V, R15). For a 3.5 kg pack, 8 g gives about 275 N and 25 g about 858 N along any axis. With a safety factor of 2, the latch proof load is about 1,717 N (1.72 kN). A steel pawl 36 mm wide with a 4 mm tooth sees about 12 MPa in shear and 36 MPa in bending at that load. Six 4 mm blind rivets in the 1.5 mm tray put about 48 MPa of bearing stress on the aluminium, a margin of about 4.2 against a conservative 200 MPa allowable, so a

doubler is prudent but not essential. To stop the pack chattering on its contacts at 8 g, a vehicle receiver preloads the pack against its end stop with at least $1.2 \times 275 = 330$ N, which needs an over-centre lever ratio of at least 6.6 for a 50 N hand force. These are paper sizings; R15 is **not verifiable at TRL 3**.

■ 7. Connector, CAN and log (R7, R10, R12)

With the assumed contact forces, insertion needs about 41 N, inside the 50 N limit of R7, but the forces depend on the contact parts, which are not yet chosen inside the decided custom keyed shroud. Two swaps a day for five years is 3,650 mating cycles, so a 5,000-cycle rating gives a margin of about 1.37.

The v0.3 message set sends about 34.1 frames per second per pack, a bus load of about 1.8 % at 250 kbit/s, or about 3.1 % with two packs on one host. The 2,000-record log is 64,000 bytes (62.5 KiB); at half the bus it transfers in about 10 s. R12 is met on paper; CSV export is dock software, not verified.

■ 8. Cost (R16)

Table 4. Cost summary from `bom/bom.csv`.

Group	Items	Cost (USD)
One pack	1 to 7, 12 to 16	436
One wall dock	8 to 11, 17	170
Total	1 to 17	606

Value-engineering target: USD 700 (`budget_usd` , a hypothetical control target, not a limit). Estimated cost of the constructable design: USD 606 (USD 94 under the target). Making the design buildable added USD 47: the latch housing, release slider and springs (line 15), the dock mounting parts (line 17), the longer back plate and more fixings (SWC-DDR-003). Cells are about \$143, or \$0.31 per Wh. Per the portfolio rule (SWC-DDR-001), this is the only place a SwapCell pack is priced.

■ 9. Results against requirements

Table 5. Requirement status. Not met: none. Items at risk and not verifiable are listed first.

ID	Value (SWC-CAL-001)	Target	Status
R9	300 to 500 cycles typical for the cell class; 4.1 V fleet mode adopted	500 cycles to 80 %; SoH within 5 %	At risk
R2	10.0 Ah and 466 Wh nominal; 9.7 Ah and 452 Wh at cell minimum	10 Ah and 450 Wh at 0.2C	At risk
R7	About 41 N insertion with assumed contact forces	10 s, one hand, 50 N or less	Not verifiable at TRL 3
R8	Gasketed lid, potted pack-side contacts (design review only)	Pack IP65, dock IP54	Not verifiable at TRL 3
R10	3,650 cycles in 5 years, margin 1.37; custom keyed shroud, contacts not yet chosen	5,000 cycles, ± 3 mm, $\pm 2^\circ$, 40 A	Not verifiable at TRL 3
R11	Short circuit 496 A, 123 A ² s in 500 μ s; pre-charge 460 ms; functions specified	Protections listed in R11	Not verifiable at TRL 3
R15	Proof load 1.72 kN; rivet bearing margin 4.2; lever ratio 6.6	Class V1 vibration and shock, no release	Not verifiable at TRL 3
R1	46.8 V nominal, 39.0 to 54.6 V	46.8 V; 39.0 to 54.6 V	Met
R3	51 °C open air, 56 °C with higher resistance; derates to 18.6 A enclosed and 14.0 A from 45 °C; 17.5 A per cell at 35 A	Cells below 60 °C from 25 °C at 20 A in an open-air mount; derate elsewhere; 35 A for 10 s	Met (on paper)
R4	2.3 h full; 1.6 h to 80 %	3 h full; 2 h to 80 %	Met
R5	3.03 kg	3.5 kg or less	Met
R6	340 x 90 x 80 mm; 393 mm overall	400 mm or less overall	Met
R12	Bus load 1.8 %; log 62.5 KiB in about 10 s	CAN 250 kbit/s, 2,000 records, CSV export	Met (on paper)
R13	Sleep drain 0.72 % per month; coded INTERLOCK at 0.30 V	Wake with no host supply; 1 % per month or less	Met (on paper)
R14	Net 3.8 A charge in the PowerBox case, within 5.0 A	Charge-discharge mode within limits	Met (on paper)
R16	USD 606 for one pack and one dock	USD 700 value-engineering target	Under the target by USD 94

■ 10. Checks against earlier documents

The TRL 2 figures in SWC-PRC-001 v0.2 were checked against this script and corrected in v0.3: mass 2.8 to 2.85 kg, grid energy 550 to 547 Wh, terminal energy 454 to 457 Wh, pack parts \$370 to \$414 (cell price raised from \$4.50 to \$5.50 and a wake button added), adiabatic rise 36 to 35 K with a better heat capacity, and 0 to 80 % charge time stated as 1.6 h. The log size is 64,000 bytes (62.5 KiB), not "64 kB" of binary kilobytes.

Constructable design (v0.3). SWC-DDR-003 made the model buildable: fixings for every part, a working latch with a release, a floating receptacle and a longer dock plate. Mass rose from 2.85 to 3.03 kg, the heat capacity from 2.26 to 2.29 kJ/K, the enclosed derating current from 18.5 to 18.6 A and the parts cost from USD 559 to USD 606. No requirement changed status. The latch sizing is unchanged: the pawl is still 36 mm wide with a 4 mm engagement, and the six 4 mm rivets now fix the latch housing, which is the doubler.

Safety: These calculations concern a 468 Wh lithium-ion pack that can deliver about 500 A into a short. They are paper estimates and do not replace protection design review or testing. No pack may be built or charged from this note; building and testing are TRL 4 work and on hold.